

Supplement S2: Reineke SDImax by Maine Ecoregion and Forest Type

Plot-specific maximum stand density index (SDImax, Reineke 1933) was estimated using a Bayesian regression model (BRMS) fit by Weiskittel and colleagues to the FIA database ($n = 173,740$ plots across the seven New England states). The model conditions SDImax on plot location, forest type, and site productivity. The 2024 posterior of plot-mean SDImax was used as the upper density envelope in the FIA CEM growth pipeline (parameter `--use_brms_sdimax`).

This supplement summarises the plot-level posterior means by ecoregion and forest-type group for the 3,253 Maine plots used in projection. Estimates are reported in metric (trees ha^{-1}) and US (trees ac^{-1}) units.

Table S2.1. SDImax by Maine ecoregion (all forest types pooled)

Ecoregion	n plots	Mean (trees ha^{-1})	Mean (trees ac^{-1})	10–90% (ha^{-1})	10–90% (ac^{-1})
Acadian Highlands	1132	1050	425	592–1563	239–633
Central Maine	761	1016	411	612–1462	248–592
Eastern/Coastal	519	982	397	567–1407	230–569
Western Mountains	841	992	401	625–1430	253–579

Table S2.2. SDImax by FIA forest-type group (Maine)

Forest-type group	n plots	Mean (ha^{-1})	Mean (ac^{-1})	10–90% (ha^{-1})	10–90% (ac^{-1})
Spruce-fir	1059	1119	453	642–1642	260–665
Softwood	255	1113	450	683–1561	276–632
Mixed	83	1061	429	649–1543	262–624
Aspen-birch	315	1027	416	564–1474	228–597
Northern hardwood	1312	935	378	596–1306	241–529
Hardwood	165	874	354	560–1285	227–520
Other	64	850	344	397–1351	161–547

Notes

Ecoregion classification follows EPA Omernik Level III, aggregated into four reporting zones used in CRSF Maine forest-condition reports: Acadian Highlands (Aroostook, Piscataquis), Western Mountains

(Franklin, Oxford, Somerset), Central Maine (Androscoggin, Cumberland, Kennebec, Penobscot, Sagadahoc, Waldo, York), and Eastern/Coastal (Hancock, Knox, Lincoln, Washington).

Forest-type groups follow the FIA FORTYPCD coarse aggregation: Spruce-fir (FORTYPCD 121–128), Softwood pine and hemlock (102–105, 167, 381), Mixed (401, 402, 409), Aspen-birch (901–904), Northern hardwood (801–809, 922), Hardwood oak/maple (503, 505, 506, 515, 519, 520, 701, 703, 704, 708), and Other.

Spruce-fir stands carry the highest density envelope (mean 1,119 trees ha⁻¹, range 642–1,642 between 10th and 90th plot-level percentiles), reflecting their narrow conical crowns and tolerance of dense self-thinning trajectories. Hardwood oak-maple stands carry the loosest envelope (874 trees ha⁻¹). Ecoregion variation in mean SDImax across all forest types is modest (982 to 1,050 trees ha⁻¹), driven mostly by differences in forest-type mixture rather than productivity per se.

These values are consumed by `06_projection_engine.R` via the `apply_sdimax_cap()` function. Projected SDI is bounded above by the plot-keyed BRMS posterior mean each cycle, and stand growth is throttled multiplicatively as relative density (SDI/SDImax) approaches 0.6 to 1.0 to honour the Reineke self-thinning ceiling.

Source

Weiskittel et al., BRMS posterior plot-level SDImax dataset, January 2024 (n = 173,740 plots, eastern US). Data archived in CRSF/FIA repository.